Pregnancy Morbidity and Mortality

Instructor: Mari-Carmen Farmer, MSN, CNM, WHNP-BC, FACNM
(Mari-Carmen.Farmer@jefferson.edu)

Block: Endocrine/Urology/Reproductive

Thread: Health Systems Science

Conflict of Interest: M.S.N. Farmer has no conflicts of interest to disclose

Learning Objectives

By exam time, you should be able to do the following:

1. Define pregnancy-related morbidity and mortality using WHO standards.
2. Describe the impact and recent trends of pregnancy-related mortality in the US.
3. Explain disparities in pregnancy-related death.
4. Identify the factors influencing pregnancy-related morbidity and mortality in the US.
5. Summarize the work being done to decrease pregnancy-related morbidity and mortality, including continued research and implementation of safety initiatives.

Study Questions

1. What is the WHO definition of pregnancy-related death?
2. What are the four most common causes of pregnancy-related death in the US, based on CDC data from 2017-2019?
3. What are some conditions that are included in pregnancy-related morbidity?
4. What are some presumed reasons that the US has higher pregnancy-related mortality rates compared to other developed countries?
5. What are some of the notable disparities in pregnancy-related mortality?

Definitions from the World Health Organization (WHO)

- **Pregnancy-related morbidity**: any health condition attributed to and/or complicating pregnancy, and childbirth that has a negative impact on well-being and/or functioning
- **Pregnancy-related death**: annual number of deaths from any cause related to or aggravated by pregnancy or its management (excluding accidental or incidental causes) during pregnancy and childbirth or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy.
- **Late pregnancy-related death**: death from direct or indirect obstetric causes, more than 42 days but less than one year after termination of pregnancy
- **Severe pregnancy-related morbidity:** accounts for a range of pregnancy-related complications that have serious short- and long-term consequences on an individual's health

Pregnancy-Related Morbidity

- This includes many conditions that may impact future well-being: hemorrhage requiring >4U blood transfusion, unplanned hysterectomy, stroke, sepsis, postop wound infections, cardiomyopathy, ARDS, pulmonary embolism, significant postpartum depression etc.
- Most of these (>70%) occur in the first two weeks after delivery
- Affects >50,000 people per year in the US; this number has nearly doubled in the past 10 years
- Readmission is very high in the 6 weeks postpartum, which causes significant stress on the patient and family, as well as significant cost to the healthcare system
- 13% of birthing people in the US report postpartum depression symptoms; suicide is the manner of death in 7.4% of pregnancy-related deaths
    - 16% of Philadelphia birthing people report postpartum depression
        - Discrepant based on race/ethnicity: Asian (20%), non-Hispanic Black (17%), non-Hispanic White (12%)
- There are significant racial and ethnic disparities in who is affected
    - From 2012 - 2015, compared with white patients, the incidence was:
        - 166% higher for Black patients
        - 148% higher for Native American and Alaskan patients
        - 122% higher for Hispanic patients
        - 117% higher for Asian/Pacific Islander patients
- Severe pregnancy-related morbidity:
    - 50,000 people per year in the US
    - 462 people per year in Philadelphia (2011-2014 data)
        - 57% higher than the national rate from 2011-2014

- List of conditions considered "severe" morbidity
  - **Disseminated intravascular coagulation**: blood clotting disorder which causes uncontrollable bleeding and bruising
  - **Renal Failure**: damage and disease to the kidneys
  - **Infection and Sepsis**: critical contamination of the body by pathogens causing disease
  - **Respiratory Distress**: difficulty breathing
  - **Eclampsia**: pregnancy-related blood pressure disorder which causes seizures
  - **Pulmonary Edema/Acute Heart Failure**: buildup of fluid in the lungs due to weakening and diminished function of the heart
  - **Thrombotic Embolism**: blood clot in a vein which can travel to organs
  - **Puerperal Cerebrovascular Disorder**: disease of blood vessels of the brain during pregnancy and the post-partum period (e.g. stroke)
  - **Amniotic Fluid Embolism**: when the fluid that surrounds the fetus in utero travels into and affects the bloodstream
  - **Hemorrhage**: loss of blood after childbirth

Pregnancy-Related Mortality

- Causes: Percentages of pregnancy-related deaths in the United States during 2017–2019:
  - Other cardiovascular conditions, 14.5%.
  - Infection or sepsis, 14.3%.
  - Cardiomyopathy, 12.1%.
  - Hemorrhage, 12.1%.
  - Thrombotic pulmonary or other embolism, 10.5%.
  - Hypertensive disorders of pregnancy, 6.3%.
  - Amniotic fluid embolism, 6.1%.
  - Cerebrovascular accidents, 5.8%.
  - Anesthesia complications, 0.2%.
  - Other non-cardiovascular medical conditions, 11.1%.

  - The cause of death is unknown for 7.0% of all 2017-2019 pregnancy-related deaths.
- Epidemiology
  - About 1205 deaths per year in the US related to pregnancy and delivery (2021)
    - In Philadelphia there are an average of 7 pregnancy-related deaths per year (2007-2018)

- An estimated 50-84% of these are preventable
- Despite high health care spending, the US has one of the highest pregnancy-related mortality rates in the developed world
  - For example, in 2020, the maternal death rate in the U.S. (17.4 per 100,000 births) was more than two times greater than Canada (8.6) and the United Kingdom (6.5).
- Pregnancy-related mortality is increasing year after year



- Causes of pregnancy-related death in the US 2017-2019

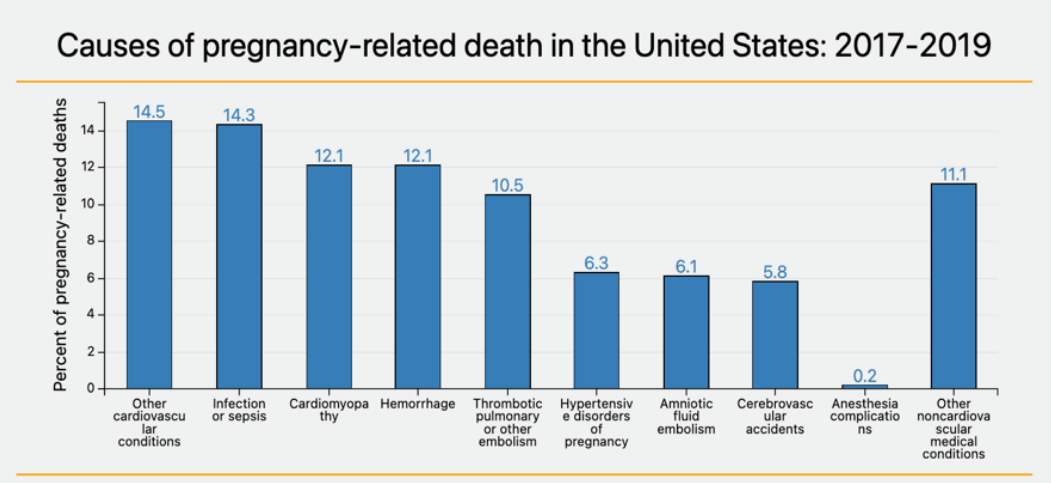
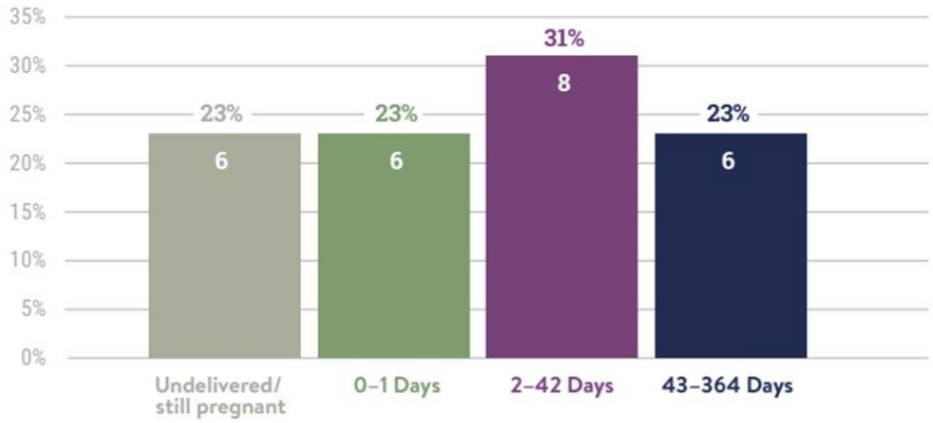


Figure 2.3 Time from End of Pregnancy Until Death for Pregnancy-Related Deaths, 2013-2018 (n=26)



- In Philadelphia, between 2013 and 2018
  - **Most pregnancy-related deaths occurred in the postpartum period**, and 23% occurred more than 6 weeks after delivery
  - Most pregnancy-related deaths were **cardiovascular related**
  - Black birthing people were **more likely** to die from a pregnancy-related cause compared to their White counterparts, irrespective of wealth or educational attainment
  - Over half (50-84%) of all pregnancy-related deaths were **preventable.**
- In 2020 there were 861 deaths (23.8/100,000) and in 2021 1205 deaths (32.9/100,000)
  - About ¼ of these were COVID related
    - direct death from COVID
    - exacerbations of social determinants (access to care, public transit, child-care, technology, living environment, employment, etc)
    - worsening of preexisting conditions
- There are significant racial and ethnic disparities in pregnancy-related death, only widened by the pandemic
  - Native Hawaiian or Pacific Islander: almost 4.5x as likely to die compared to non-hispanic white people
  - Non-hispanic Black: 2.8x
  - Native American: 2.3x

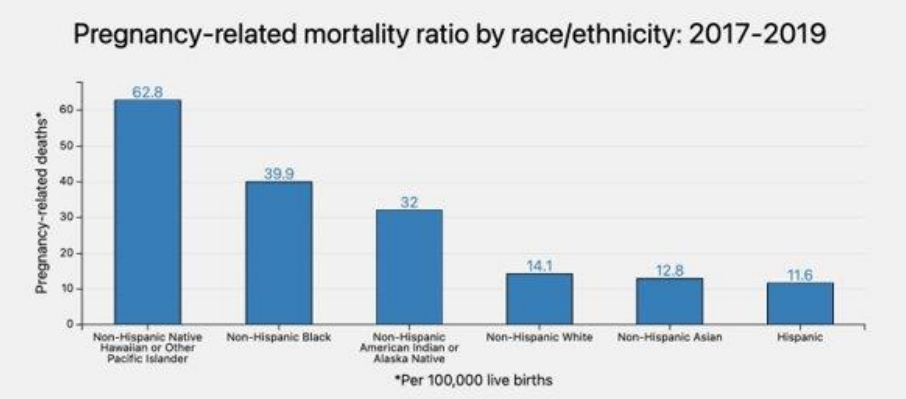
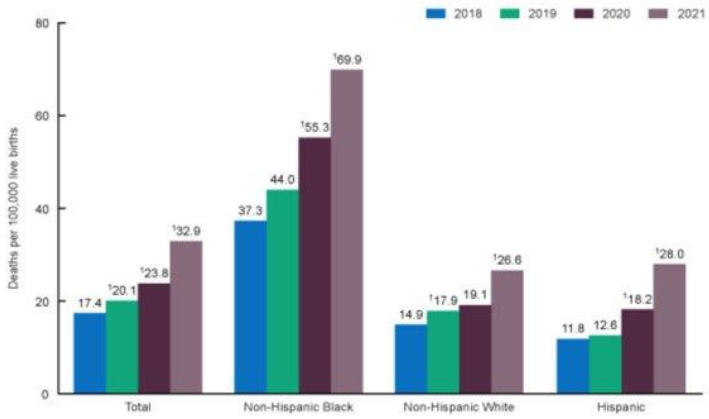
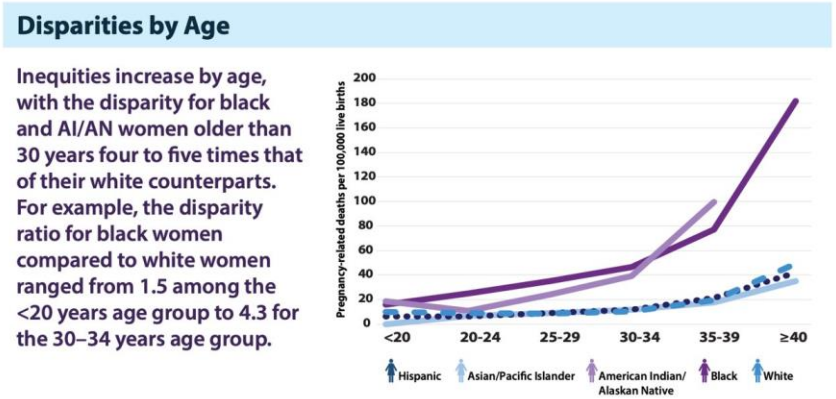


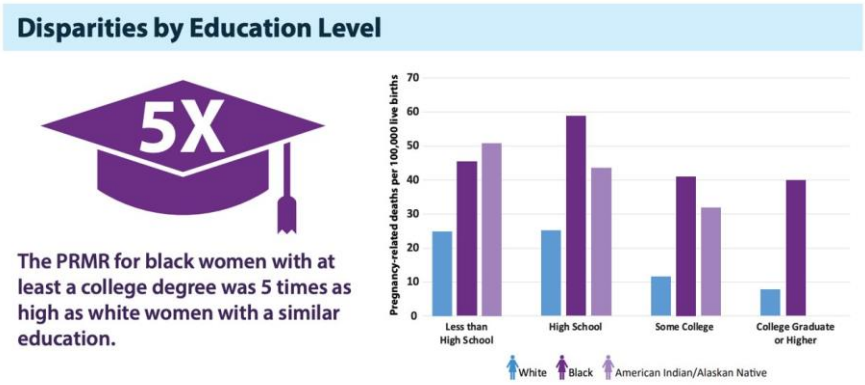
Figure 1. Maternal mortality rates, by race and Hispanic origin: United States, 2018–2021



- There are also disparities by age, with those disparities being much more significant for black patients and Native American patients

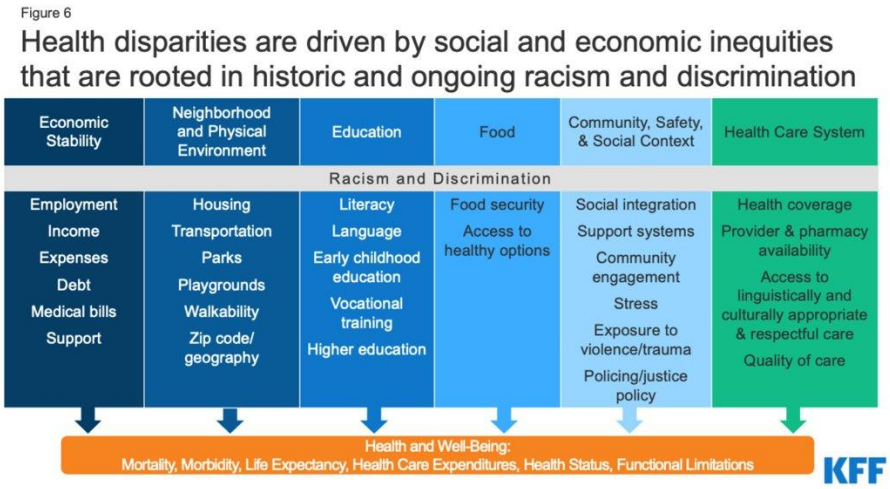


- Those racial and ethnic disparities continue to hold true when compared by education level



Why is the US pregnancy-related mortality rate so high and so disparate?

- Pre-pregnancy health
  - Primary and preventative care less central to our health system
  - Not enough primary care providers and lack of access in many parts of the country
  - Increasing prevalence of preexisting conditions such as hypertension, diabetes, and obesity
  - Worsening mental health population wide
  - Increased opioid use, overdose, and suicide
    - Opioid-related diagnoses at the time of delivery increased by 131% between 2010 and 2017
- Pregnancy care
  - Shortage of pregnancy care providers and lack of access to care
    - Other countries have higher number of midwives compared to their pregnant populations
    - Closing L&D units nationwide due to decreased profitablity, often in areas where access is already a problem (small hospitals and facilities)
    - In 2019, 76.7% of pregnant people received adequate and early prenatal care--> people who do not receive prenatal care are 3-4 x more likely to die from pregnancy related complications
- Post-birth care
  - Most developed countries have access to home visits, paid parental leave, access to ongoing healthcare
  - Less access to family and community networks
- Health disparities deeply affect pre-pregnancy health, pregnancy care and post-birth care



Next steps: what work is being done and needs continued work?

- Research
  - Many studies ongoing- no need to memorize any of them
  - See the NIH web portal for examples (in PowerPoint slides)
- Work on "safety bundles"
  - The Alliance for Innovation on Maternal Health (an ACOG initiative) has been developing safety bundles
  - These bundles have the goal to support best practices and make birth safer
  - Topics include obstetric hemorrhage, severe hypertension in pregnancy, reducing primary cesarean, cardiac conditions in pregnancy, care for patients with substance use disorder, perinatal mental health, postpartum care, and sepsis in obstetrics
- Distribution of care providers and hospitals
  - Regionalized health care systems can reduce pregnancy-related morbidity and mortality
    - Stabilize and transfer
    - Affiliation with academic/higher resource hospitals
  - Increased access to midwifery care
- Access to safe abortion
  - Abortion restricted states have increased pregnancy-related death (fewer providers, less access to care, disproportionately affecting patients with fewer resources)
- Access to prenatal care and health care in the year following pregnancy
- Access to care BEFORE pregnancy
- Individual and workforce/systems work on implicit bias and anti-racism
- Diversification of the pregnancy care and primary care workforce

Answers to Study Questions

Q: What is the WHO definition of pregnancy-related death?

A: Pregnancy-related deaths, according to the WHO, include the annual deaths from any cause related to or aggravated by pregnancy or its management (excluding accidental or incidental causes) during pregnancy and childbirth or within 42 days of termination of pregnancy, irrespective of the duration and site of the pregnancy. A "late pregnancy-related death" in one that occurs more than 42 days but less than 1 year after the end of a pregnancy (regardless of duration of pregnancy).

Q: What are the four most common causes of pregnancy-related death in the US, based on CDC data from 2017-2019?

A: The four most common causes of pregnancy-related death based on CDC data from that period (pre-COVID) include "other" cardiovascular conditions (this includes MI, valvular disease, coronary dissection, endocarditis, etc.), infection/sepsis, cardiomyopathy, and hemorrhage.

Q: What are some conditions that are included in pregnancy-related morbidity?

A: Conditions such as hemorrhage requiring >4 units of blood transfused, unplanned hysterectomy, eclampsia, stroke, sepsis, cardiomyopathy, ARDS, significant postpartum depression, and postoperative wound infections are included in this category.

Q: What are some presumed reasons that the US has higher pregnancy-related mortality rates compared to other developed countries?

A: Primary/preventative care plays a less significant role in our health system, leading to more health concerns prior to pregnancy. Fewer pregnancy-care providers per pregnant person as other developed countries so less access to care. People who do not get prenatal care are 3-4x more likely to die from a pregnancy-related complication. Disparities in health care access and delivery cause significant increases in pregnancy-related deaths. Mental health has significantly worsened, and opioid-related diagnoses, overdoses, and suicide have increased.

Q: What are some of the notable disparities in pregnancy-related mortality?

A: Compared to non-Hispanic white patients, Native Hawaiian or Pacific Islander patients are almost 4.5x as likely to die, non-Hispanic black patients are 2.8x as likely to die, and Native American patients are 2.3x as likely to die from conditions related to pregnancy. Mortality rates have continued to rise during the pandemic, with those increases also being disparate. The increase in death rate by age shows that black and Native American patients older than 30 years of age are 4 to 5 times as likely to die from a pregnancy-related complication than their white, age-matched counterparts. Black patients with a college degree or higher are also 5 times more likely to die from a pregnancy-related complication than white patients with a similar education.